

TURUKHANSK, Russian Federation

WMO: 234720

Lat: 65.794N

Lon: 87.989E

Elev: 124

StdP: 14.63

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-52.1	-46.6	-56.0	0.2	-52.0	-51.0	0.3	-46.5	21.0	4.4	19.5	1.3	3.2	60	0.752

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	81.2	64.4	77.2	62.7	73.3	61.0	66.9	76.9	64.6	73.7	62.5	70.7	6.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	87.2	71.2	61.1	80.8	68.5	58.7	74.1	66.7	31.6	77.1	29.9	73.7	28.3	70.9	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.5	14.4	DB	-54.4	85.6	6.2	4.0	-58.8	88.4	-62.4	90.8	-65.9	93.0	-70.4	95.9	(4)
(5)			WB	-54.4	69.4	6.2	2.8	-58.8	71.4	-62.4	73.1	-65.9	74.7	-70.3	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.1	-15.7	-9.2	7.5	20.3	34.9	54.8	61.2	54.6	42.3	24.2	-2.7	-9.8	(6)
(7)		DBStd	29.81	18.47	15.49	15.38	13.06	9.97	10.18	7.36	7.19	8.25	11.77	18.44	18.57	(7)
(8)		HDD50	10974	2036	1657	1318	890	481	67	4	31	258	800	1580	1852	(8)
(9)		HDD65	15757	2501	2077	1783	1340	935	331	164	331	683	1265	2030	2317	(9)
(10)		CDD50	774	0	0	0	0	12	210	351	175	26	0	0	0	(10)
(11)		CDD65	82	0	0	0	0	0	25	47	10	0	0	0	0	(11)
(12)		CDH74	705	0	0	0	0	3	234	409	59	0	0	0	0	(12)
(13)		CDH80	142	0	0	0	0	0	48	84	10	0	0	0	0	(13)
(14)	Wind	WSAvg	7.0	7.3	7.0	7.4	7.5	7.1	6.6	6.2	6.3	6.7	7.6	6.8	7.6	(14)
(15)	Precipitation	PrecAvg	23.7	1.3	1.2	1.3	1.3	1.7	2.2	2.4	3.2	2.6	3.0	1.9	1.7	(15)
(16)		PrecMax	31.3	2.8	2.6	3.0	2.6	3.3	5.7	4.8	5.1	7.5	4.9	3.5	3.5	(16)
(17)		PrecMin	16.7	0.4	0.2	0.2	0.3	0.6	0.4	0.1	0.8	1.0	1.7	0.6	0.4	(17)
(18)		PrecStd	4.2	0.6	0.6	0.6	0.7	0.7	1.3	1.2	1.3	1.5	1.0	0.7	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.5	26.6	39.1	52.7	71.0	86.0	86.4	80.5	69.8	50.9	34.8	29.0	(19)
(20)			MCWB	24.6	25.5	32.3	40.7	54.3	65.3	67.3	66.5	56.5	47.2	33.1	28.0	(20)
(21)		2%	DB	17.1	21.4	34.0	46.3	60.3	80.6	82.1	74.4	62.4	44.1	31.9	25.7	(21)
(22)			MCWB	15.7	20.8	30.7	37.7	47.5	62.3	65.2	62.3	53.0	41.6	31.1	24.8	(22)
(23)		5%	DB	12.9	16.3	30.5	42.0	54.5	76.2	78.7	70.1	57.9	40.9	28.2	21.3	(23)
(24)			MCWB	12.0	15.2	28.0	35.3	44.3	60.6	63.8	60.3	51.3	38.8	27.3	20.3	(24)
(25)		10%	DB	8.5	12.4	27.8	38.2	48.9	71.7	75.0	66.2	54.2	37.7	23.7	15.8	(25)
(26)			MCWB	7.6	11.3	25.9	33.1	40.8	58.5	62.6	58.9	49.8	35.5	22.7	14.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.8	25.6	33.4	41.2	54.6	68.2	70.7	68.2	58.7	47.5	33.8	28.4	(27)
(28)			MCD B	25.6	26.2	36.5	49.9	67.9	80.5	81.9	78.2	65.8	51.3	34.4	28.8	(28)
(29)		2%	WB	16.1	20.8	31.1	38.4	48.7	65.0	67.9	64.4	55.4	42.4	31.0	24.8	(29)
(30)			MCD B	16.9	21.5	33.8	44.8	58.5	76.4	77.3	70.6	60.9	43.6	31.8	25.7	(30)
(31)		5%	WB	12.1	15.3	28.4	36.1	45.2	62.3	65.8	61.9	52.7	39.1	27.3	20.4	(31)
(32)			MCD B	13.0	16.1	30.1	40.7	53.3	73.1	75.3	67.9	56.6	40.6	28.3	21.2	(32)
(33)		10%	WB	7.6	11.4	25.8	33.8	41.5	59.9	63.9	59.5	49.9	35.6	22.7	14.8	(33)
(34)			MCD B	8.4	12.4	27.8	38.1	47.8	69.7	72.5	65.1	54.1	37.2	23.8	15.8	(34)
(35)	Mean Daily Temperature Range			MDBR	12.3	14.5	17.7	17.9	13.7	15.6	15.7	12.7	10.2	8.5	12.4	(35)
(36)		5% DB	MCDBR	12.8	16.7	13.5	15.0	19.6	22.2	22.0	18.5	15.8	8.2	10.3	13.4	(36)
(37)			MCWBR	12.7	16.5	11.4	10.1	11.4	10.2	10.0	9.1	9.2	7.1	10.3	13.6	(37)
(38)			MCD BR	13.0	16.9	13.3	13.6	18.2	18.9	18.8	15.6	13.0	8.1	10.4	13.1	(38)
(39)		5% WB	MCWBR	12.9	16.6	11.9	10.0	11.4	10.0	9.9	9.2	8.8	7.5	10.5	13.2	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.224	0.275	0.319	0.342	0.363	0.380	0.415	0.399	0.342	0.283	0.208	0.220	(40)
(41)		taud		1.858	1.985	1.994	2.036	2.157	2.387	2.272	2.288	2.402	2.174	1.891	1.866	(41)
(42)		Ebn at Noon		61	176	226	255	264	263	247	238	229	177	64	7	(42)
(43)		Edh at Noon		7	20	33	40	40	33	36	32	23	16	6	2	(43)
(44)	All-Sky Solar Radiation	RadAvg		31	230	705	1327	1690	1739	1566	994	562	212	61	7	(44)
(45)		RadStd		4	20	67	148	157	148	189	123	74	39	9	0	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.76	N/A	N/A	N/A	N/A	N/A	-670	-819	N/A	N/A
(47) Regional (2 neighbors)		+1.76	N/A	N/A	N/A	N/A	N/A	-670	-819	N/A	N/A

Nomenclature: See separate page